

Model Report: Transformer Encoder (SITS)

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Script: /home/mmann1123/Documents/github/South_Africa_Crop_Comp/deep_learn/src/Deep Learning/Pixel_Field_Level

Data Split:

train_count: 4802658
val_count: 1205790
test_count: 1531914
random_seed: 42
split_method: fid-wise

Metrics (test set):

Accuracy: 0.8337
Cohen Kappa: 0.7656
F1 Weighted: 0.8320
F1 Macro: 0.7810

Training Time: 5802s (96.7 min)

Hyperparameters:

architecture: Transformer encoder (multi-head self-attention, mean-pooled)
d_model: 128
n_head: 8
n_layers: 3
dim_feedforward: 256
dropout: 0.3
lr: 0.001
optimizer: AdamW (wd=1e-4)
scheduler: CosineAnnealingLR
loss: WeightedFocalLoss(gamma=2.0, alpha=1/class_counts)
model_selection: val F1 macro (maximize)
gradient_clipping: 1.0
epochs: 100
patience: 15
batch_size: 2048
n_models: 5
seeds: [42, 101, 202, 303, 404]
temporal_order: chronological
positional_encoding: sinusoidal (actual month positions)
pooling: temporal mean
norm: pre-norm (norm_first=True)

Per-Class Metrics

Class	Precision	Recall	F1-Score	Support
Barley	0.752	0.757	0.754	140
Canola	0.904	0.879	0.891	107
Lucerne/Medics	0.918	0.945	0.931	380
Small grain grazing	0.716	0.578	0.640	83
Wheat	0.669	0.708	0.688	120

Confusion Matrix

